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The Effect of Sulfate-Rich Water Production on Surface Production Lines: A Lab Study

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Abstract

Calcium sulfate (CaSO_4) scale precipitation can cause significant operating problems during oil production. Impermeable hard scale deposits of calcium sulfate can severely impair production in various locations, such as the formation matrix, well tubulars, and surface production lines. The development of calcium sulfate mitigation protocols is crucial to an effective flow assurance strategy, which should include prevention, characterization, and removal. This study shows an effective chemical formulation that has been proven to restore production and protect equipment in a time-efficient manner. The study illustrates three stages of scaling tendency study including precipitation and deposition, simulation, and removal.

Two synthetic water samples were prepared to represent water from different sources, one with high calcium ions (CRW), while the other high in sulfate ions (SRW). Static bottle tests were conducted to replicate scale formation and deposition. XRD and ESEM analyses were conducted on the generated scale sample to confirm the composition. An extensive sensitivity and simulation analysis were conducted on many water samples with different ranges of calcium and sulfate ions to determine the ion concentration precipitation threshold. The effect of saturated gases, namely CO_2 , was also studied in terms of precipitated compounds. Finally, dissolution tests were conducted using variety of chemical fluid combinations and conditions to determine the optimal dissolution fluid.

The results showed mixing samples associated with high calcium and sulfate ions can cause not only calcium sulfate but calcite precipitation as well. The calcium sulfate scale was found to adhere to test tube walls more strongly than the pure calcite or calcium sulfate. More importantly, the treatment of this scale is much more difficult, highlighting the necessity of applying mitigation measures when these scales are anticipated. The dissolution tests showed that manipulation of temperature, soaking time, and types of dissolvers are important factors for higher dissolution success. A proposed solution was prepared based on green chemicals to achieve around 90% dissolution capacity of calcite/calcium sulfate mixture.

This paper presents the methodology used to determine the source of scale, factors that influence the formation of scale, and shows how calcite/ CaSO_4 scale can be removed using effective chemistry and removal protocols. Details of this study can be expanded to cater to many formation damage incidents faced in the oil and gas industry.

Introduction

Calcium sulfate (CaSO_4), a naturally occurring mineral composed of calcium, sulfur, and oxygen, is commonly found in various geological formations. In oil production operations, calcium sulfate scale can form when calcium-rich formation water interacts with sulfate-containing injected fluids, such as seawater. This phenomenon can occur in downhole environments—such as within the formation matrix and production tubing—as well as in surface equipment like flowlines and separators (Paudyal et al., 2020; Alhamad, 2020). Calcium sulfate scaling can cause significant operational challenges, including flow restrictions, equipment failure, and increased maintenance costs (Delorey et al., 1996; Fan et al., 2010; Alhamad, 2020; Paudyal et al. 2020; Alotaibi et al., 2023).

The formation of calcium sulfate scale is primarily driven by the mixing of incompatible waters. For instance, seawater injection introduces high concentrations of sulfate ions into reservoirs that may already contain calcium-rich brines. The resulting chemical imbalance promotes calcium sulfate precipitation, particularly under varying conditions of temperature, pressure, and pH (Delorey et al., 1996; Paudyal et al., 2020). Additionally, scaling may occur during water production from commingled zones or during pressure drawdown.

During matrix acidizing, carbonate formations are typically treated with hydrochloric acid to dissolve the rock and create conductive channels for hydrocarbon flow. However, calcium sulfate, often present as a discrete mineral layer or near natural fractures, can react with the acid treatment fluids. When the acid contacts calcium sulfate, sulfate ions dissolve and mix with the calcium ions released from the carbonate. As the acid is spent and the pH increases, the system can reach supersaturation, triggering calcium sulfate precipitation within the etched pathways. If the acid contacts sulfate-bearing minerals before reaching the carbonate zone, early scale nucleation may occur, reducing the efficiency of the stimulation (Frenier and Ziauddin, 2008; Kan et al., 2009; Paudyal et al., 2020).

Calcium sulfate exists in multiple forms, including anhydrite (CaSO_4), gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$), and hemihydrate ($\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$). Each form has distinct solubility and thermal stability characteristics. Anhydrite, for instance, is less soluble and stable at high temperatures, whereas gypsum is more common in surface or shallow subsurface conditions. The specific form that precipitates depends on reservoir conditions, such as temperature and fluid composition, and directly impacts the effectiveness of treatment and mitigation strategies (Foroutan and Moghadasi 2013; Alotaibi et al., 2023).

Mechanical methods, including scraping, jetting, and milling, are often used for scale removal, but these are time-consuming and expensive, especially in deep or deviated wells. Chemical treatments using hydrochloric acid or specialized dissolvers offer an alternative but are often limited by the low solubility of calcium sulfate, particularly anhydrite. Acid treatments may also cause corrosion or secondary precipitation if not carefully managed (Crabtree et al., 1999; Frenier and Ziauddin 2008).

Inhibition offers a preventive strategy through continuous or batch injection of scale inhibitors, which can chelate calcium ions or modify crystal growth. These inhibitors must be tailored to well-specific conditions, including ionic strength, temperature, and flow regime. Common application methods include continuous injection through capillary strings or squeeze treatments into the formation. Proper selection and deployment of inhibitors are critical for maintaining long-term system integrity and production efficiency (Smith et al., 1968; Kan et al., 2009; Shen and Crosby 1983).

Despite existing solutions, challenges remain in understanding calcium sulfate precipitation under downhole conditions. This study investigates calcium sulfate scale formation and removal in oil production environments, focusing on the evaluation of chemical formulations and treatment effectiveness under simulated field conditions.

Materials

This study utilized lab-prepared synthetic water samples representing two different sources, each with distinct geochemical characteristics. The first water source, referred to as "Calcium-Rich Water" (CRW), was prepared to have a high concentration of calcium ions, ranging from 10,000 to 40,000 ppm. The second water source, referred to as "Sulfate-Rich Water" (SRW), was prepared to have a high concentration of sulfate ions, exceeding 1,000 ppm. The geochemical analysis of these two water sources is presented in [Table 1](#). Dissolved iron was added to the samples to account for production tubular corrosion byproducts leaching into the water. Low amounts of typical downhole gas impurities were also bubbled through the water samples to mimic more realistic conditions.

Table 1—Geochemical Analysis of the Two Prepared Water Samples

Geochemical components	Unit	SRW	CRW
Bicarbonate	mg/L	800 – 1,200	200 – 400
Calcium		1,500 – 2,500	30,000 – 40,000
Chloride		35,000 – 45,000	120,000 – 130,000
Magnesium		1,500 – 2,500	5,000 – 6,000
Potassium		1,500 – 2,500	1,500 – 2,500
Sodium		20,000 – 25,000	25,000 – 30,000
Strontium		50 – 60	8,000 – 1,200
Sulfate		5,500 – 6,500	8,000 – 1,200
pH	-	6.5 – 7.5	4 – 5
Specific Gravity	g/L	1.02 – 1.08	1.10 – 1.15

Compatibility Test

To assess the compatibility of the CRW and SRW, a series of mixing tests at 200°F for 24 hours were conducted to evaluate the potential for scale formation. The CRW and SRW were mixed in different volume ratios, including 5:95, 25:75, 50:50, and 75:25 (CRW:SRW), to evaluate the effect of varying water chemistry on scale formation.

Upon mixing the two waters, a black coloration was immediately observed, which is due to the presence of hydrogen sulfide (H_2S) in the SRW, as can be seen in [Figure 1](#). The black coloration is likely due to the formation of iron sulfide (FeS) or other sulfur-containing compounds, which can react with the iron present in the water to form a black precipitate.

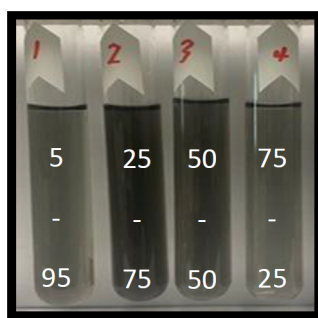


Figure 1—Observed black coloration due to the formation of iron sulfide (FeS) upon mixing the two waters

After allowing the mixed waters to settle, a scale was observed to form, which exhibited a reddish color, as can be seen in Figures 2 and 3. The reddish coloration is consistent with the presence of magnetite (Fe_3O_4). The X-Ray Diffraction (XRD) analysis of the scale, given in Table 2, revealed that it was primarily composed of gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$), with traces of magnetite (Fe_3O_4) and halite (NaCl).

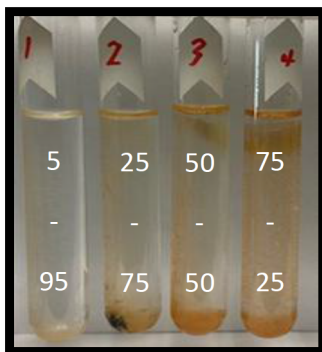


Figure 2—Observed precipitation from mixing CRW and SRW at 200°F



Figure 3—Collected scale from compatibility test of mixing CRW and SRW

Table 2—XRD results of the precipitated salts from mixing CRW and SRW

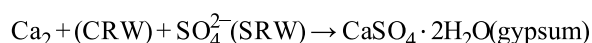
Compound	Precipitation (Wt%)
Gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$)	90
Calcite (CaCO_3)	8
Aragonite (CaCO_3)	1
Magnetite (Fe_3O_4)	Traces
Halite (NaCl)	Traces

The Environmental Scanning Electron Microscopy (ESEM) analysis of the scale provided further insight into its composition, with the results shown in Table 3. The ESEM results indicated the scale is primarily composed of oxygen (36%), iron (18%), chlorine (15%), and calcium (15%), with smaller amounts of magnesium, sulfur, sodium, and silicon. The high oxygen content is consistent with the presence of gypsum, which has a molecular formula of $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$. The iron content is attributed to the presence of magnetite.

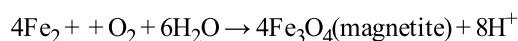
Table 3—ESEM Analysis of Scale Composition

Element	Precipitation (Wt%)
O	36
Fe	18
Cl	15
Ca	15
Mg	1
S	9
Na	5
Si	0.9

The formation of gypsum in the mixed water sample can be explained by the following chemical reaction:



This reaction occurs when the calcium ions in the CRW react with the sulfate ions in the SRW, resulting in the formation of gypsum. The presence of magnetite in the scale can be attributed to the oxidation of ferrous iron (Fe^{2+}) in the water, which reacts with oxygen to form magnetite:



The presence of halite in the scale is likely due to the high salinity of the mixed water sample, which can lead to the precipitation of sodium chloride (NaCl) as a secondary mineral phase.

The results of the compatibility tests demonstrate that the mixing of CRW and SRW can lead to the formation of a scale, that is primarily composed of gypsum, with traces of magnetite and halite. The presence of hydrogen sulfide and iron sulfide in the mixed waters can result in the formation of a black precipitate, while the scale itself exhibits a reddish color due to the presence of magnetite. These findings have important implications for the management of scaling in oilfield operations, and highlight the need for careful consideration of water chemistry and compatibility in the design of water treatment and injection systems.

Sensitivity Study

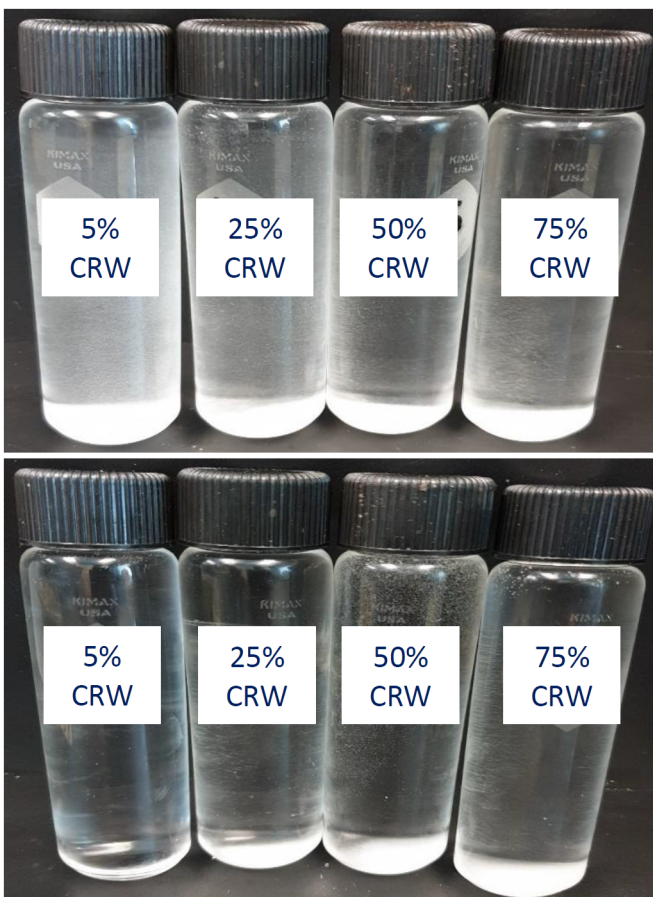
To further investigate the compatibility of the CRW and SRW, a series of sensitivity tests were conducted. The tests involved varying the concentration of calcium in the CRW and sulfate in the SRW, and evaluating the effect on the precipitation of calcium sulfate and calcium carbonate.

The concentration of calcium in the CRW was varied between 17,000 ppm and 38,000 ppm, while the concentration of sulfate in the SRW was varied between 1,000 ppm and 6,000 ppm. Table 4 shows the different combinations of calcium and sulfate concentrations that were tested.

Table 4—Variation in Calcium and Sulfate in CRW and SRW, Respectively

CRW Calcium Concentration	SRW Sulfate Concentration
Max (38,000 ppm)	Max (6,000 ppm)
Max (38,000 ppm)	Min (1,000 ppm)
Min (17,000 ppm)	Min (1,000 ppm)
Min (17,000 ppm)	Max (6,000 ppm)

For each combination of calcium and sulfate concentrations, compatibility tests were conducted at different ratios of CRW to SRW, ranging from 1 vol% CRW to 75 vol% CRW. The results of the compatibility tests for the most severe case (Max calcium and sulfate) showed that both calcium sulfate and calcium carbonate precipitated in all of the cases. Figure 4 shows pictures of some tubes with deposited scale. To distinguish between the two precipitates, a few droplets of hydrochloric acid (HCl) were added to the testing tubes. The HCl dissolved the calcium carbonate, leaving behind the calcium sulfate.



After aging:

Test tubes showing scale precipitation at different percentages of CRW produced water to SRW produced water.

After adding HCl:

Same test tubes after adding few droplets of HCl were added showing reduction in scale (calcium carbonate was dissolved) while remaining solids were assumed to be calcium sulfate.

Figure 4—Pictures of the testing tubes for the most severe case

The thresholds or amounts of CRW (ratio) needed to initiate precipitation of calcium sulfate were determined for each combination of calcium and sulfate concentrations. The results are presented in Table 5.

Table 5—Thresholds of examined combinations to get calcium sulfate precipitation

CRW Calcium Concentration	SRW Sulfate Concentration	Threshold CRW Ratio
Max (38,000 ppm)	Max (6,000 ppm)	1 vol%
Max (38,000 ppm)	Min (1,000 ppm)	22.5 vol%
Min (17,000 ppm)	Min (1,000 ppm)	No scale
Min (17,000 ppm)	Max (6,000 ppm)	1 vol%

● **Effect of Maximum Concentrations:**

Max CRW (38,000 ppm) & Max SRW (6,000 ppm): Precipitation begins at just 1% CRW. This indicates that when both calcium and sulfate levels are at their highest, even a minimal amount

of CRW can trigger scaling, highlighting the heightened propensity for precipitation under these conditions.

- **Impact of Varying Sulfate Levels:**

Max CRW (38,000 ppm) & Min SRW (1,000 ppm): Precipitation starts at 22.5% CRW. This shows that even with lower sulfate levels, high calcium concentrations can still lead to precipitation, though a higher proportion of CRW is required.

- **Influence of High Sulfate with Lower Calcium:**

Min CRW (17,000 ppm) & Max SRW (6,000 ppm): Precipitation starts at 1% CRW. This indicates that high sulfate levels, even with lower calcium concentrations, can still lead to rapid scaling.

The results of the sensitivity analysis show that the precipitation of calcium sulfate is highly dependent on the concentrations of calcium and sulfate in the CRW and SRW, respectively. The threshold CRW ratio required to initiate precipitation of calcium sulfate varies significantly depending on the combination of calcium and sulfate concentrations.

In general, the results suggest that higher concentrations of calcium and sulfate increase the likelihood of calcium sulfate precipitation, and lower CRW ratios are required to initiate precipitation. However, the exact threshold CRW ratio required to initiate precipitation of calcium sulfate depends on the specific combination of calcium and sulfate concentrations.

Simulation

To further investigate the effects of temperature and pressure on the precipitation of gypsum and anhydrite, a simulation was conducted using a commercial software. The simulation was run at a 1:1 ratio between the two waters, with the goal of predicting the impact of varying mixing conditions on the fluid-to-fluid compatibility between the SRW and the CRW.

The simulation was conducted over a range of pressures (14.7 – 400 psi) and temperatures (80 – 160°F), based on the Pitzer theory of electrolytes. The Pitzer theory is a widely used approach for modeling the behavior of electrolyte solutions, and it considers the interactions between ions and the solvent.

The simulation results were evaluated in terms of the scaling index (SI) predictions, which provide an indication of the potential for inorganic scale precipitation. A positive SI value ($SI > 0$) indicates a supersaturated solution, and there is a potential for inorganic scale precipitation. On the other hand, a negative SI value ($SI < 0$) indicates an under-saturated solution, and no scale precipitation would be encountered.

The simulation results showed that the varying temperatures had a negligible impact on gypsum, with no significant changes in the SI values observed over the range of temperatures studied, as can be seen in [Figure 5](#). In contrast, a slight increase in the saturation index of anhydrite was observed with increasing temperature, as given in [Figure 6](#). This suggests that temperature may play a role in the precipitation of these scales, although the effect is relatively small.

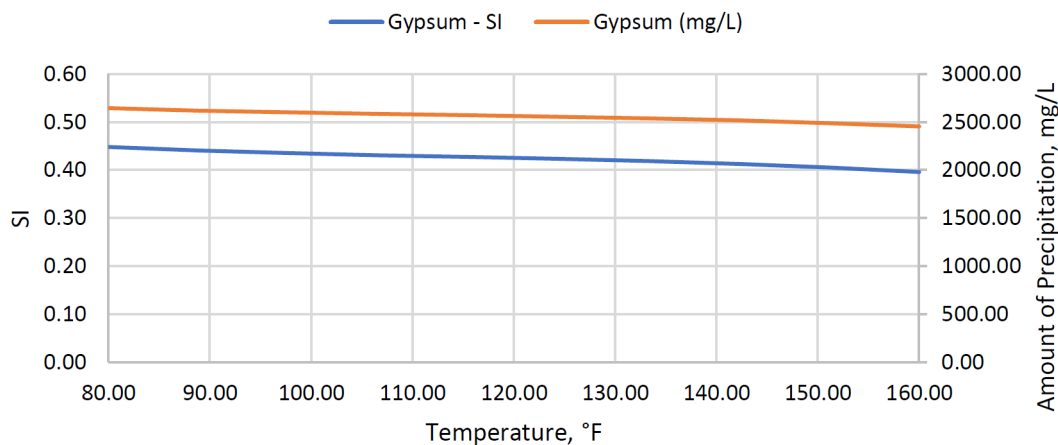


Figure 5—Temperature effect in the precipitation of gypsum

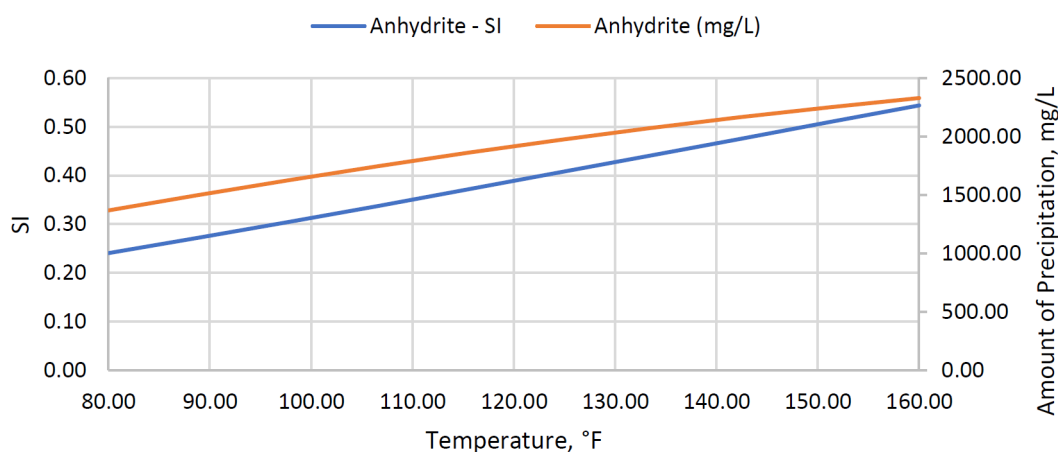


Figure 6—Temperature effect in the precipitation of anhydrite

The simulation results also showed that pressure had no noticeable impact on the scaling indices of gypsum and anhydrite. Figures 7 and 8 show the SI plots versus pressure changes for both gypsum and anhydrite scales. This suggests that pressure is not a significant factor in the precipitation of these scales, at least over the range of pressures studied.

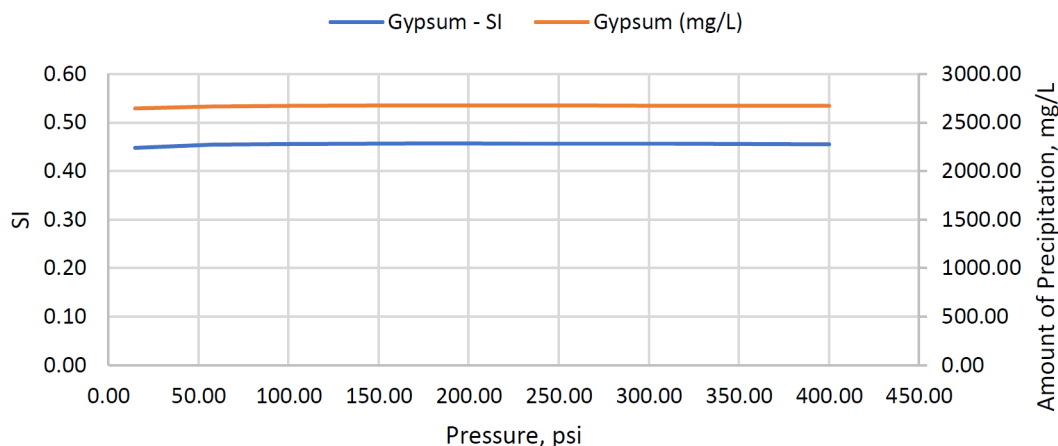


Figure 7—Pressure effect in the precipitation of gypsum

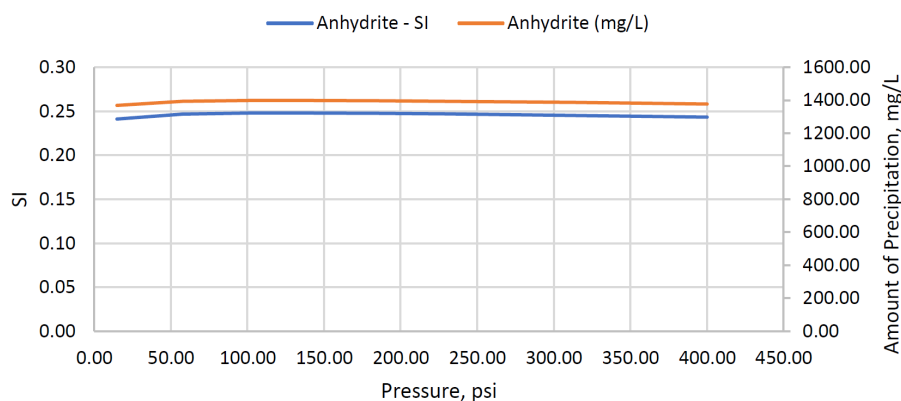


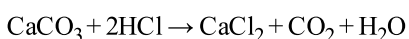
Figure 8—Pressure effect in the precipitation of anhydrite

The results of the simulation study have important implications for the management of scaling in oilfield operations. The fact that temperature has a negligible impact on gypsum precipitation suggests that temperature control may not be an effective strategy for preventing gypsum scaling. On the other hand, the slight increase in the saturation index of anhydrite with increasing temperature suggests that temperature control may be an effective strategy for preventing these scales.

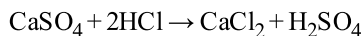
The fact that pressure has no noticeable impact on the scaling indices of gypsum, and anhydrite suggests that pressure control may not be an effective strategy for preventing these scales. This is consistent with the results of previous studies, which have shown that pressure has a limited impact on the precipitation of these scales.

Solubility and Dissolution of the Scale

The solubility of calcium sulfate in HCl and chelating agents is an important consideration in oilfield operations, as it can affect the potential for successful treatments. HCl is usually the first choice of acidizing due to its low cost and availability (Alhamad et al., 2021). Calcium carbonate and calcium sulfate are both very common scales in the oil industry. However, it is very important to understand the different solubility limit between calcium sulfate and calcium carbonate in HCl. Calcium carbonate is highly soluble in HCl. The reaction between calcium carbonate (CaCO_3) and HCl is:



This reaction is highly favorable, and calcium carbonate readily dissolves in HCl, releasing carbon dioxide gas. The solubility of calcium carbonate in HCl is essentially unlimited, meaning it will completely dissolve in the acid. Calcium sulfate, on the other hand, is less soluble in HCl (Delorey et al., 1996; Mahmoud et al., 2021; Alhamad et al., 2023). While it will still react with the acid, the reaction is not as favorable as with calcium carbonate:



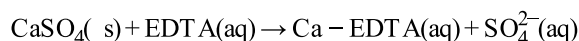
In this reaction, the HCl molecule donates a proton (H^+) to the calcium sulfate, which forms a soluble calcium chloride complex. The sulfate ion (SO_4^{2-}) is released into the solution, where it can react with the hydrogen ion (H^+) to form sulfuric acid (H_2SO_4).

The solubility of calcium sulfate in HCl is limited. At room temperature, the solubility of CaSO_4 in 1M HCl is approximately 2 – 3 g/L at 20°C. This means that calcium sulfate will only partially dissolve in HCl, and the amount of dissolution will depend on the concentration of the acid and the temperature.

Chelating agents are molecules that can form multiple bonds with a metal ion, effectively "grabbing" onto it and making it more soluble, stable, and bioavailable. The term "chelate" comes from the Greek word "chele," meaning "claw," which describes the way these molecules wrap around and bind to metal

ions. Chelating agents work by forming a complex with a metal ion, which involves the donation of one or more pairs of electrons to the metal ion, increasing its solubility, reducing its reactivity, and enhancing its bioavailability (Al-khaldi et al., 2011; Almubarak et al., 2017; Alotaibi et al., 2023). Common examples of chelating agents include EDTA, citric acid, and oxalic acid, which have applications in various industries such as water treatment, food and beverage processing, pharmaceuticals, and environmental remediation, making them an essential tool for managing metal ions and improving their properties.

Calcium carbonate and sulfate can be dissolved by chelating agents that can complex with the calcium ion (Ca^{2+}) (Shaughnessy and Kline, 1983; Cikeš et al., 1990; Al-Khaldi et al., 2011; Almubarak et al., 2017). The solubility of calcium sulfate in chelating agents, such as EDTA, can be evaluated using the following equation:



The solubility of calcium sulfate in EDTA is relatively high, with a solubility of approximately 500 g/L at 20°C. The solubility of calcium sulfate in EDTA increases with increasing EDTA concentration. In this reaction, the EDTA molecule binds to the calcium ion, forming a stable and soluble chelate complex. The sulfate ion (SO_4^{2-}) is released into the solution, where it can react with other ions to form a precipitate.

The mechanism of solubility of calcium sulfate in EDTA is more complex than in HCl, involving multiple steps and the formation of a stable and soluble chelate complex. The mechanism of solubility of calcium sulfate in HCl is simpler, involving a single acid-base reaction.

In this study, the dissolution tests were conducted to evaluate the effectiveness of various chemicals in dissolving the calcium sulfate scale. The tests involved weighing a scale sample piece (2 to 5 grams) and soaking it in a glass bottle with different chemicals for a specified period of time. The sample was dried and weighed after each stage, and the total solubility was calculated using the weight loss method:

$$\text{Solubility}(\text{wt}\%) = (\text{InitialWeight} - \text{FinalWeight}) / (\text{InitialWeight}) \times 100$$

The dissolution tests were designed to optimize the parameters that affect the dissolution efficiency of the calcium sulfate scale. These parameters included temperature, fluid stages/sequences, dissolver to solid ratio, and repetition of stages.

Initially, water showed a limited dissolution of the calcium sulfate scale, with a solubility of 7.7% at 50°C. In contrast, hydrochloric acid (HCl) showed a significantly higher dissolution rate, with a solubility of 22% at 50°C. The EDTA-based fluid showed the highest dissolution rate, with a solubility of 40% at 50°C.

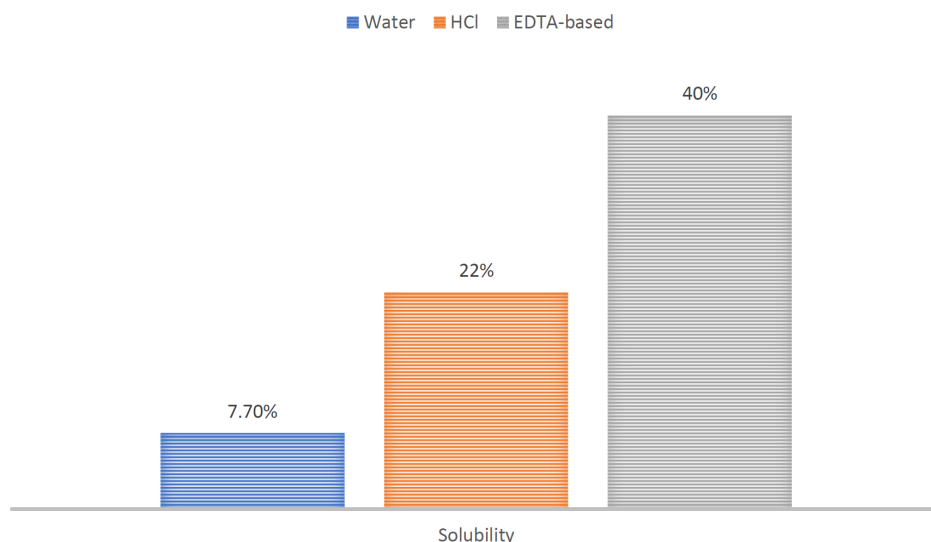


Figure 9—Solubility of calcium sulfate in water, 15 wt% HCl and EDTA-based fluid

Sulfate ions were also traced in these solutions to confirm solubility of calcium sulfate and the results agreed with the weight loss results as can be seen in Figure 10.

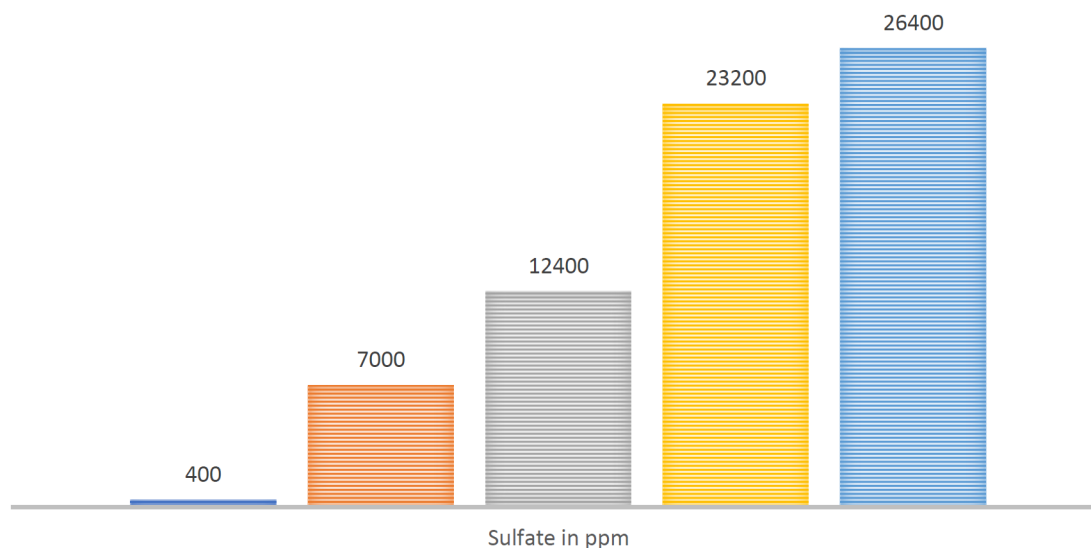


Figure 10—Dissolved sulfate ions in ppm in different solutions

The EDTA-based fluid was further optimized by varying the temperature and applying the fluid multiple times sequentially. The results showed that increasing the temperature and repeating the fluid application significantly improved the dissolution rate. At 80°F, the dissolution rate increased to 54% after 2 days and 61% after 4 days. At 120°F, the dissolution rate increased to 75% after 2 days. Noting that a fresh fluid sample was used every 24 hours.

With respect to soaking time, the results showed that increasing the soaking time significantly improved the dissolution rate. At 120°F, the dissolution rate increased to 40% after 16 hours, 52% after 1 day, 75% after 2 days, 83% after 3 days, 90% after 4 days, and 94% after 5 days.

Table 6—Dissolution of Different Dissolvers for Calcium Sulfate Sample

Fluid	Temperature (°C)	Dissolution (%)
Water	50	7.7
HCl		22
EDTA-Based	50	40
	80	54 (2 days)
		61 (4 days)
	120	75 (2 days)
		94 (5 days)

The chemical analysis of the dissolution tests suggests that the EDTA-based fluid is the most effective in dissolving the calcium sulfate scale. The EDTA molecule is a chelating agent that binds to the calcium ions, forming a stable and soluble complex. This complexation reaction can be represented by the following equation:



The formation of the Ca-EDTA complex is a critical step in the dissolution process, as it allows the calcium ions to be removed from the scale and dissolved in the fluid.

The results of the dissolution tests also suggest that temperature and time are critical parameters in the dissolution process. Increasing the temperature and soaking time significantly improved the dissolution rate, indicating that the reaction is kinetically controlled. The results of the dissolution tests suggest that the activation energy for the dissolution reaction is relatively low, indicating that the reaction is facile and can occur at relatively low temperatures.

The practical implications of these results are that the EDTA-based dissolver offers a highly effective method for dissolving calcium sulfate scale, particularly in industrial settings where scale buildup can lead to equipment damage and operational inefficiencies. However, considerations for real-world applications include the cost of the EDTA-based dissolver compared to HCl and potential environmental impacts, which would need to be evaluated alongside its effectiveness.

Conclusions

The study investigated the formation, inhibition, and removal of calcium sulfate scale in oil production systems. The results showed that the mixing of CRW and SRW can lead to the formation of a scale that is primarily composed of gypsum, with traces of magnetite and halite. The sensitivity analysis revealed that the precipitation of calcium sulfate is highly dependent on the concentration of calcium and sulfate ions in the CRW and SRW.

The simulation study showed that temperature and pressure have a negligible impact on the precipitation of gypsum, while the saturation index of anhydrite increases with increasing temperature. The dissolution tests showed that EDTA-based fluid is the most effective in dissolving calcium sulfate scale, with a dissolution rate of up to 90% after 4 days. The results also showed that temperature and time are critical parameters in the dissolution process.

The study concluded that the formation of calcium sulfate scale can be prevented or removed using effective chemical formulations. The results of the study have important implications for the management of scaling in oilfield operations and highlight the need for careful consideration of water chemistry and compatibility in the design of water treatment and injection systems. The study also demonstrated the effectiveness of EDTA-based fluid in dissolving calcium sulfate scale, which can be used in industrial settings to prevent equipment damage and operational inefficiencies.

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